

Samyukta Suryanarayanan

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EDUCATION

Columbia University, Columbia Engineering

New York, NY

M.S. in Computer Science | Machine Learning Track | May 2026

Rutgers University Honors College

Camden, NJ

B.S. in Computer Science | Minors: Business Administration, Mathematics | Outstanding Achievement Scholarship

TECHNICAL SKILLS

Programming: Python, SQL, Java, C++, JavaScript

Agentic AI & LLMs: Multi-agent orchestration, LangChain, LlamaIndex, FastMCP, Google Genkit, Prompt Engineering, Trace-based Grounding, Structured Output Validation, Hallucination Mitigation

RAG & Retrieval: FAISS, Amazon Bedrock, LangChain retrieval/reranking, document understanding, Precision@k · Recall@k · MRR · NDCG

Data Engineering: ETL pipelines, Celery, PostgreSQL, AWS (S3, SQS, RDS, EC2, Lambda, Bedrock), DynamoDB, Redis

ML & Full-Stack: PyTorch, Scikit-learn, XGBoost, MLflow, React 18, Next.js, Django 5.0, Flask, FastAPI, Streamlit

Evals & Testing: pytest, LangSmith, promptfoo, CI/CD via GitHub Actions, Docker, Structured JSON output validation

EXPERIENCE

AI Engineer (Part-Time) | HerbShield, Remote

Jan 2025 – Present

- Built the core AI matching system using Google Genkit (TypeScript), powering personalized herbal remedy recommendations based on symptoms, severity, and care type, with Zod schema validation enforcing structured output integrity across all flows.
- Developed a WhatsApp chatbot delivering AI-personalized messaging throughout the full customer journey, from program signup through herbal medication reminders and progress updates, and built the website chatbot so prospective customers can ask questions about the program and get instant answers.
- Core team member on a small founding team, contributing to all product decisions including database schema design, program packaging, and website architecture; designed and built the full Next.js / Firebase / Stripe web application end-to-end.

Software Engineer (Part-Time) | CyberGhostOps, Remote

Jan 2025 – Present

- Set up and managed the GitHub repository for a Flask-based cybersecurity education platform, establishing version control workflows, branch structure, and deployment processes to support iterative development.
- Architected a Claude AI integration via Model Context Protocol (MCP), connecting the website to Claude to automate content updates and site management workflows, replacing manual processes with LLM-powered automation.

AI/ML Technical Consultant & Curriculum Engineer | Independent

May 2024 – Present

- Designed and delivered end-to-end ML engineering curriculum (Python, PyTorch, Scikit-learn, ETL pipelines) for 10+ learners; advised on ML pipeline failures, model validation, and production-readiness using the same documentation and testing standards applied in enterprise AI delivery.

PROJECTS

Multi-Agent AI Security Auditor (SSRF Auditor)

- Designed and shipped a 3-agent LLM orchestration harness (URL Analyst, Sanitization Checker, Judge) combined with Semgrep static analysis; ran 9 ablation conditions on 120 real-CVE files, lifting recall from 38% (static-only) to 86%, with every architecture decision grounded in EvidencePacket production traces rather than intuition.
- Confirmed trace-based grounding as load-bearing via controlled ablation (removal degraded recall and increased false negatives); built pytest eval harness enforcing structured JSON output validation across all agent boundaries; achieved 50%+ automated remediation with evaluation reports aligned to enterprise AI requirements (LangSmith / promptfoo).

AI-Powered Domain-Specific Insights Assistant (RAG System)

- Built end-to-end RAG pipeline on AWS Bedrock with per-domain FAISS vector indexing, LangChain retrieval chain with page-level document citations (PDF/DOCX/CSV/JSON/TXT), and structured JSON output (answer, key_insights[], confidence), covering document understanding across heterogeneous enterprise corpora.
- Implemented Precision@k, Recall@k, MRR, and NDCG@k from scratch wired into a ground-truth benchmark runner; eval suite directly drove retrieval architecture, chunking, and reranking decisions, with zero regressions via offline pytest suite across 3 modules.